Geophys. J. Int. (1998) **133,** 44–56

Palaeomagnetism of the Lower Ordovician Orthoceras Limestone, St. Petersburg, and a revised drift history for Baltica in the early Palaeozoic

Mark A. Smethurst,[1] Alexey N. Khramov[2] and Sergei Pisarevsky[2]

[1] *Geological Survey of Norway,* PO Box 3006, *Lade,* N-7002 *Trondheim, Norway. E-mail: mark.smethurst@ngu.no*
[2] *All-Russian Petroleum Research and Geological Prospecting Institute, Liteiny av.* 39, *St. Petersburg* 191104, *Russia*

Accepted 1997 October 2. Received 1997 October 1; in original form 1997 May 15

SUMMARY

Palaeomagnetic investigation of Lower Ordovician limestone in the vicinity of St. Petersburg yields a pole position at latitude 34.7°N, longitude 59.1°E ($dp/dm = 5.7°/6.4°$). A probable primary remanence origin is supported by the presence of a field reversal. The limestone carries one other remanent magnetization component associated with a Mesozoic remagnetization event.

An apparent polar wander path is compiled for Baltica including the new result, ranging in age from Vendian to Cretaceous. Ages of the published Lower to mid-Palaeozoic palaeomagnetic pole positions are adjusted in accordance with the timescale of Tucker & McKerrow (1995). The new Arenig result is the oldest of a series of Ordovician and Silurian palaeomagnetic pole positions from limestones in the Baltic region. There are no data to constrain apparent polar wander for the Tremadoc, Cambrian and latest Vendian. If the Fen Complex results, previously taken to be Vendian in age (*c.* 565 Ma), are reinterpreted as Permian remagnetizations, an Early Ordovician–Cambrian–Vendian cusp in the polar wander path for Baltica is eliminated. The apparent polar wander curve might then traverse directly from poles for Vendian dykes on the Kola peninsula (*c.* 580 Ma) towards our new Arenig pole (*c.* 480 Ma). The consequence of this change in terms of the motion of Baltica in Cambrian times is to reduce significantly a rotational component of movement.

The new Arenig pole extends knowledge of Ordovician apparent polar wander an increment back in time and confirms the palaeolatitude and orientation of Baltica in some published palaeogeographies. Exclusion of the Fen Complex result places Baltica in mid- to high southerly latitudes at the dawn of the Palaeozoic, consistent with faunal and sedimentological evidence but at variance with some earlier palaeomagnetic reconstructions.

Key words: APW (apparent polar wander), limestone, Ordovician, palaeomagnetism, St. Petersburg.

1 INTRODUCTION

Our understanding of the palaeolatitudinal motion and rotation history of Baltica though Palaeozoic time has increased enormously during the 1990s. Torsvik *et al.* (1990) produced an apparent polar wander (APW) path for Baltica which prompted a concerted effort to define the course of APW better (Torsvik *et al.* 1992, 1996; Smethurst 1991; Trench & Torsvik 1991). At present the most conspicuous break in the palaeomagnetic record for Palaeozoic Baltica is between late Vendian and Lower Ordovician time, lasting ≈80 Myr. The present palaeomagnetic study, in Tremadoc, Arenig and Llanvirn age rocks, aimed to help fill this gap in APW and to constrain better the early Palaeozoic palaeogeography of Baltica.

The Lower Ordovician sequence in the St. Petersburg region consists of 10–20 m of limestones, sandstones and mudstones (Figs 1 and 2). The succession is part of a 2 km thick Vendian to Devonian cover sequence resting unconformably on Archaean and Proterozoic crystalline basement rocks. The cover rocks dip slightly to the south and southeast to form the northwestern limb of the Moscow syncline. Local neotectonic structures are of glacial origin.

Limestones dominate the Lower Ordovician sequence, giving rise to a low escarpment running from the southern shores of

© 1998 RAS

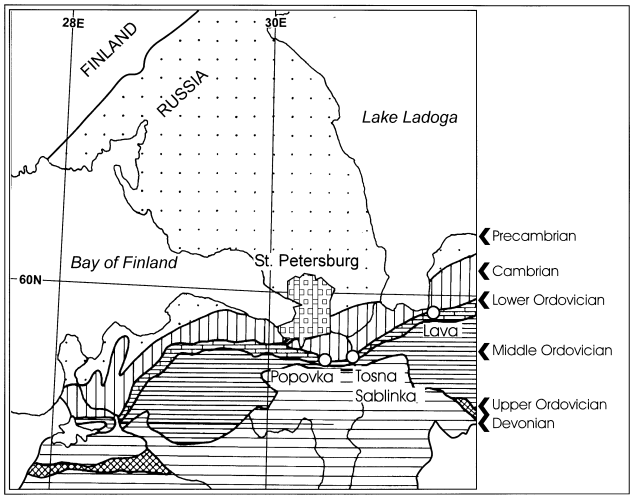


Figure 1. Geological map of the St. Petersburg region, Russia. Popovka, Tosna/Sablinka and Lava are sampling localities. Dots: undifferentiated Precambrian rocks; vertical lines: Cambrian; blocks: Lower Ordovician; narrow horizontal lines: Middle Ordovician; diagonal hatching: Upper Ordovician; wide horizontal lines: Devonian.

the Bay of Finland to Lake Ladoga (Fig. 1). They become increasingly dolomitized and richer in glauconite down sequence. Local oxidation of the glauconite gives rise to red, yellow, brown and violet blotches and layers.

The limestones are rich in well-preserved fossils, including trilobites (*Asaphys* and *Megalaspis*), cephalopods (*Endoceras* and *Cylendoceras*) and graptolites, a faunal assemblage indicative of a warm open sea palaeoenvironment (Selivanova & Kofman 1971). Using mainly the graptolite fauna as a guide, Obut, Sennikov & Dmitrovskaya (1991) inferred a Tremadoc age for the sandy limestones at the base of the sequence and an Arenig age for the remainder (Fig. 2). K/Ar ages for glauconite grains from the Arenig part of the sequence range between 431 and 512 Ma (Shukolukov & Bibikova 1986).

The Ordovician rocks lie essentially undisturbed on the Baltic craton—the 'core' of the Palaeozoic palaeocontinent of Baltica. They have remained within a few hundred metres of the surface for their entire history. Several stratigraphic gaps exist above them; missing strata include the Early Silurian, the Early Devonian, the Early and Late Carboniferous, the Early Permian and the Early Triassic (Selivanova & Kofman 1971).

The first palaeomagnetic study of the Ordovician strata was carried out by one of us (ANK) in 1958, in the early days of palaeomagnetic investigation. A dual-polarity NRM was reported which corresponded to a palaeomagnetic pole at 34°N, 135°E (pole number 11010, Khramov 1984). Thermal and alternating field 'cleaning' did little to shed light on the multicomponent structure of the remanent magnetization (Khramov & Rodionov 1977).

2 PALAEOMAGNETIC RESULTS

We sampled the Lower Ordovician sequence where the Popovka, Tosna, Sablinka and Lava rivers incise the limestone escarpment south and east of St. Petersburg (Fig. 3). In all, 59 oriented block samples were collected in eight partially overlapping traverses through the sequence (Fig. 2). A total of 278 25 mm diameter cylindrical specimens were extracted from the block samples.

Measurements of natural remanent magnetization (NRM) were carried out in the palaeomagnetics laboratory at the Geological Survey of Norway using a JR5 spinner magnetometer. Supplementary measurements were made using more sensitive 2G cryogenic magnetometers at the Lamont-Doherty Earth Observatory of Columbia University and at the Department of Geological Sciences, University of Michigan. Initial NRM intensities were invariably low; 80 per cent of them below 1 mA m^{-1}. 258 of the cylindrical specimens were progressively demagnetized. Thermal demagnetization was mainly used, the alternating field (AF) method proving less effective.

The specimens behaved in one of three ways during demagnetization: (1) they demagnetized steadily up to 680 °C to reveal a single vector component of NRM; (2) two oblique remanence components were revealed, the most stable being removed by treatments up to 580 °C; or (3) noisy behaviour during demagnetization obscured the original NRM. The first two types of behaviour are described below.

2.1 Specimens carrying largely single-component NRM

At Popovka the NRM was dominated by a remanence component removed by laboratory treatments between 200° and 680 °C, directed towards the southwest, pointing upwards (Figs 4a and b). The remanence component unblocked gradually up to ≈500 °C, where demagnetization almost ceased. Demagnetization continued between 540° and 650 °C, randomizing the remaining remanence to leave the specimens demagnetized by ≈675 °C.

A similar remanence structure was observed in a significant proportion of specimens from the nearby Tosna and Sablinka river sections. Many specimens from the more distant Lava River section also carried single-component remanence, similar in every way to the other results, except that the direction of the component was opposite to that in the other river sections (Figs 4c and d). This component, therefore, has dual polarity, with the polarities separated spatially rather than stratigraphically. We argue for a secondary, Mesozoic, origin for this component and therefore refer to it as S. The occurrences of the two S polarities in the stratigraphy are shown in Fig. 2, where reverse polarity is indicated by −S.

We notice that the light-coloured limestone, sometimes with brick-red patches, is the predominant carrier of S. The 'peach' colour indicates chemical alteration of glauconite to iron oxides such as goethite, limonite and haematite. Goethite may carry part of the remanence S, with unblocking temperatures up to the order of 200°–300 °C. The thermally stable part of S, with a narrow unblocking temperature range between 550° and 680 °C, is probably carried by haematite, the narrow unblocking temperature spectrum suggesting haematite in specular form. The presence of high-coercivity phases is borne out by isothermal remanent magnetization (IRM) experiments, which showed that saturation is not achieved by 1.2 T, and coercivities of remanence (H_{cr}) exceed 0.6 T. It is therefore concluded that the principal carrier of S is haematite.

2.2 Specimens carrying multicomponent NRM

A low-stability remanence component pointing steeply downwards was identified in all limestone types in the Tosna/Sablinka

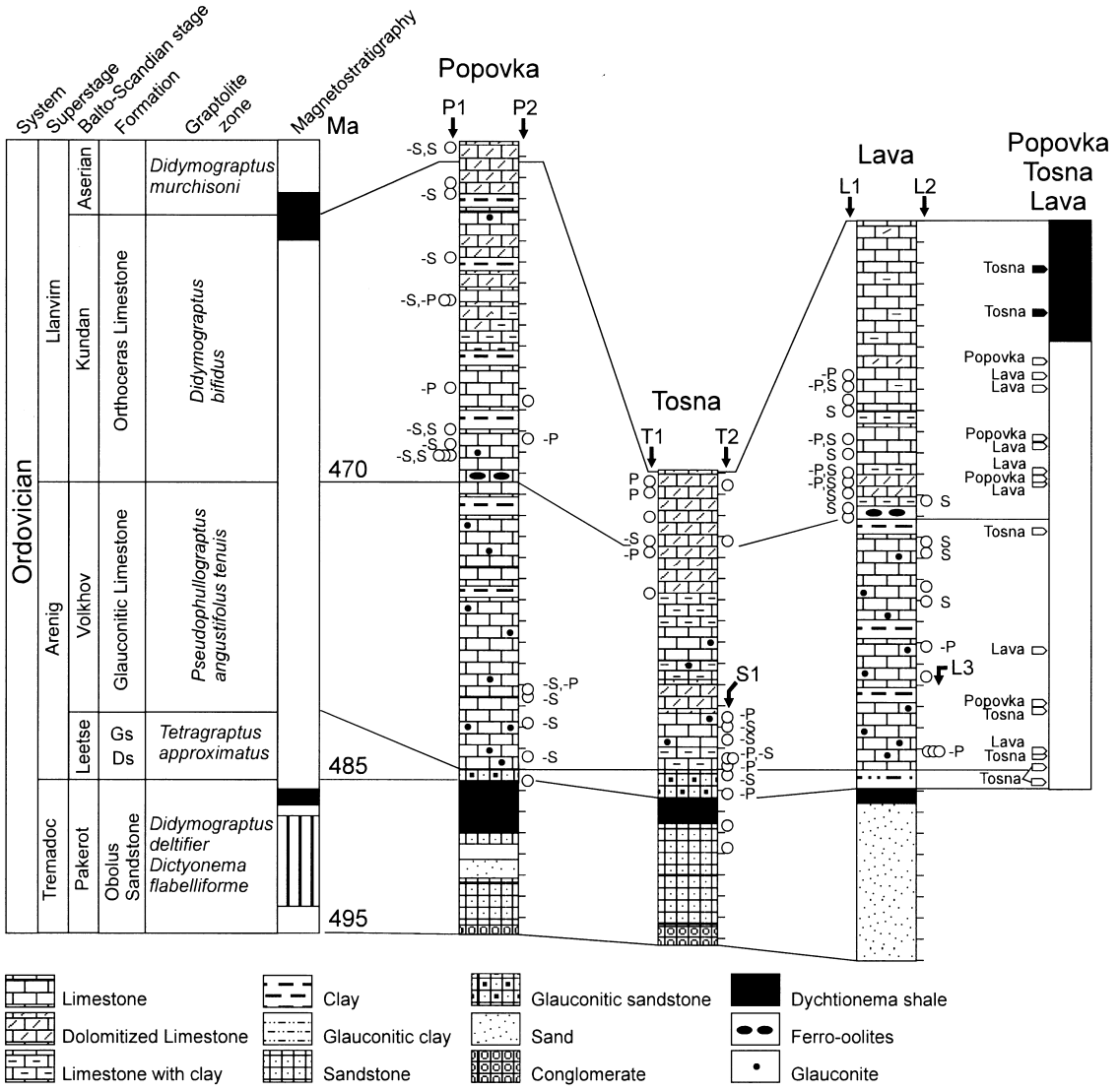


Figure 2. Stratigraphic logs for the Lower Ordovician rocks at Popovka, Tosna/Sablinka and Lava. Left: stratigraphic nomenclature and timescale according to Tucker & McKerrow (1995). The reference magnetostratigraphic log is after Trench *et al.* (1991), which is a revision of the log of Khramov *et al.* (1965). Centre: the stratigraphic logs; P1, P2, T1, T2, S1, L1, L2 denote sampling traverses at Popovka, Tosna, Sablinka and Lava (Fig. 3); circles denote stratigraphic positions of block samples; P (S) indicates primary (secondary) remanence components; −P and −S denote reverse-polarity components; ticks on the logs are at 0.5 m intervals. Right: magnetostratigraphic log based on remanence component P from the three stratigraphic sections; black (white) symbols indicate occurrences of normal (reverse) polarities in the studied sequences. Linear interpolation was used to project results from stratigraphic units with variable thickness into the composite magnetostratigraphic log.

and Lava river sections (Fig. 5). Most instances of this remanence component were near-vertical, the rest steep and northerly, parallel to the present geomagnetic field in St. Petersburg. The near-vertical population of directions is presumed to have some causative link with drilling of cylindrical specimens from the original block samples. Since the most readily oriented rock surfaces in the field were horizontal bedding planes, drilling was carried out directly into the oriented surfaces, parallel to the vertical axis of the blocks, to minimize composite orientation errors. It is therefore possible that drilling might have steepened up already steep present Earth's field directions carried by the limestones. Whatever the details of their origin, we presume that the low-stability component is of recent age and has no bearing on the Ordovician palaeofield.

In the glauconitic limestones demagnetization of the steep remanence component was followed either by noisy behaviour or by the removal of a second, oblique, remanence component. The second remanence was demagnetized between $\approx 300\,^{\circ}C$ (20 mT) and 590 °C (100 mT), and was directed either to the southeast and down (Figs 5a and b) or to the northwest and up (Fig. 5c). We argue for a primary, Early Ordovician mode for this dual-polarity remanence, and therefore refer to it as P. We established the actual polarity of P using a reference APW path for Baltica defined below; normal and reverse occurrences of P are shown in the stratigraphic sections of Fig. 2. Although the remanence component is identified in all river sections, it is best represented in the Tosna/Sablinka section, where S is poorly developed. Normal-polarity P directions were only identified in the youngest strata in the Tosna River section.

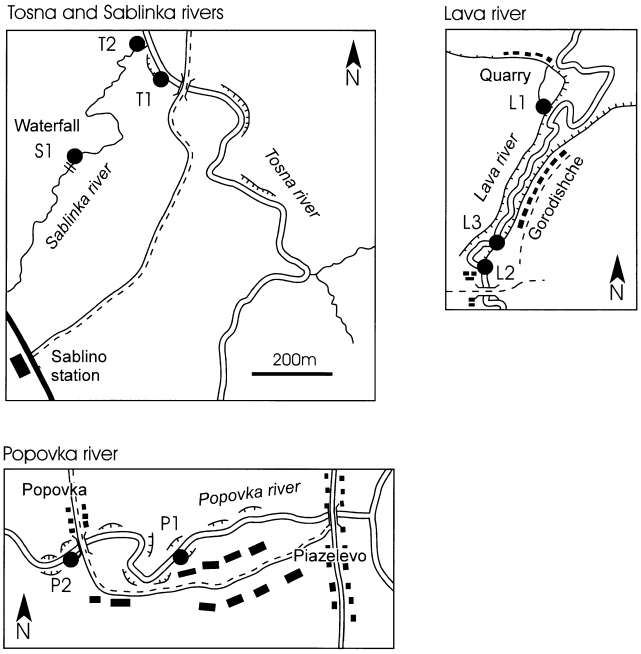


Figure 3. The locations of sampling traverses within the three sampling regions of Fig. 1. Top left: traverses T1, T2 and S1 along the Tosna and Sablinka rivers; top right: traverses L1, L2 and L3 along the Lava River; bottom: traverses P1 and P2 on the Popovka River. The scale bar on the top left map applies to all sections.

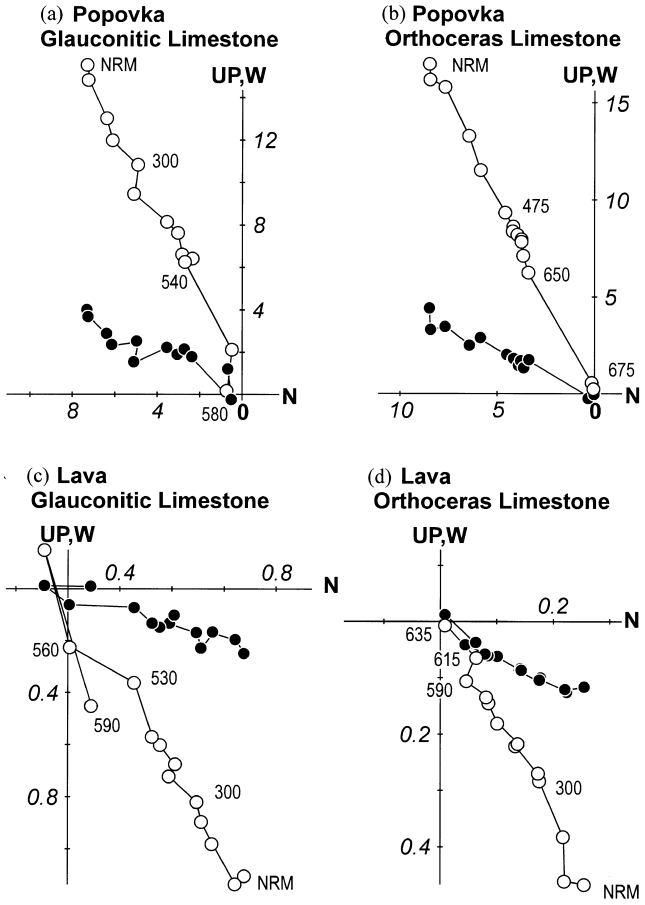


Figure 4. Orthogonal plots of progressive thermal demagnetization data showing magnetizations interpreted to be secondary remanence components (S) of Mesozoic age. Closed (open) symbols denote data projected onto the horizontal (vertical) plane. Temperatures are in °C. (a), (b) Almost all instances of S in the Popovka River section were of reverse polarity. (c), (d) S was exclusively of normal polarity in the Lava River section. The few S remanence components seen in Tosna/Sablinka samples were of reverse polarity, consistent with the nearby Popovka River section (Fig. 1).

Many of the specimens carrying the multicomponent remanence structure described above were green, containing diagenetic glauconite. Some specimens include high-coercivity minerals; saturation of IRM in these cases was not achieved by 1.2 T, and H_{cr} exceeded 0.4 T. Traces of a lower-coercivity mineral were indicated by a slight levelling off of some of the IRM acquisition curves at 0.1–0.2 T. Other specimens saturated rapidly, almost completely by 0.4 T. In these cases H_{cr} was of the order of 0.1 to 0.2 T, indicating the presence of lower-coercivity minerals.

The high-coercivity phase is taken to be goethite, which is presumed to carry the low thermal stability remanence component (haematite remanences were not in evidence in the demagnetization data). The lower-coercivity phase is taken to be magnetite, which is presumed to carry the remanence component P. P was demagnetized between 300 °C (20 mT) and 590 °C (100 mT), close to the Curie point of magnetite.

2.3 The magnetization mode of S

All S remanence directions are given in Fig. 6(a) ($n = 113$). The normal-polarity group ($n = 49$), predominantly from Lava, is tightly clustered. The wider scatter in reverse-polarity S directions is partly due to distortion of the strata around a neotectonic fold/push-up structure at Popovka, where local dips reach 20°, and partly due to noisy demagnetization data from specimens at other localities. If we only consider the clean data from Popovka, taken from the limbs of the neotectonic fold, we are left with 28 of the original 64 reverse-polarity S directions (Fig. 6a, grey diamonds). Tilt 'correction' has little effect on the majority of S directions, given that dips of strata seldom exceed 2° (Fig. 6b). The cluster of 28 selected directions from the fold structure at Popovka, however, tightens significantly upon unfolding of the structure ($P = 0.05$, McFadden & Jones 1981; Fig. 6 grey diamonds, Table 1). We take these 28 unfolded directions to be representative of the reverse-polarity palaeofield S. We take the tight group of 49 normal-polarity S directions to be significant in unfolded form.

The mean directions for the normal- and reverse-polarity counterparts of S are only 6° away from exact anti-polarity (Table 1, Fig. 7). This exceeds the critical angle of 4° for the reversal test of McFadden & McElhinny (1990) and therefore the data fail the test ($P = 0.05$). We suspect, however, that a portion of the 6° deviation may be due to slight inaccuracies in the tilt correction of the data, and we combine the two polarities into a single mean direction (Table 1). Despite its dual polarity, S is interpreted to be secondary because the polarities vary with geographic location rather than stratigraphic position (Fig. 2). A secondary mode for S is consistent with the nature of its mineral host. Rock magnetic properties suggest that some combination of haematite and goethite carries S, perhaps derived from the chemical breakdown of early diagenetic glauconite. The dual polarity of S indicates two geographically separate phases of chemical remagnetization.

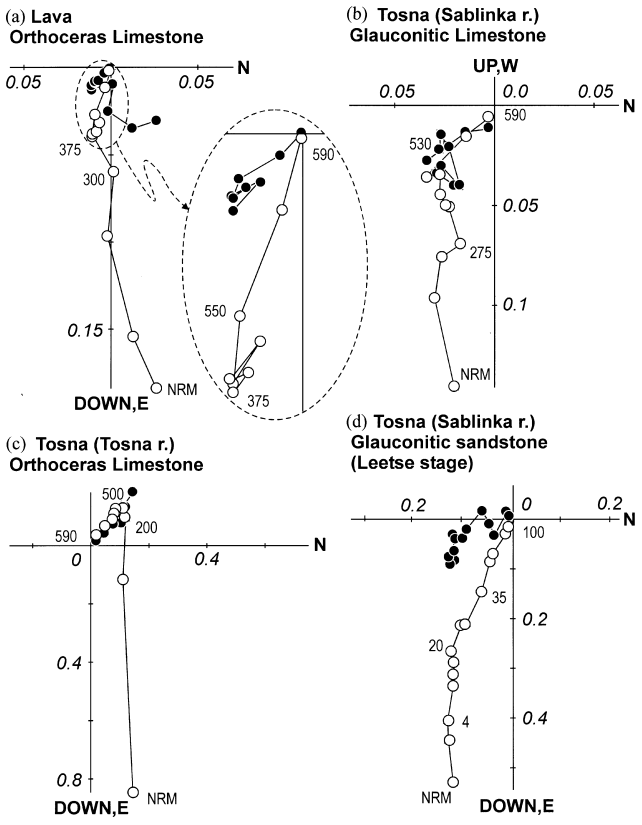


Figure 5. Orthogonal plots of thermal (a, b, c) and AF (d) demagnetization data showing magnetizations interpreted to be primary remanence components (P) of Early Ordovician age. All specimens carrying P also carry a low-stability downward-directed remanence component which is usually vertical, sometimes northerly. (a), (b) P in reverse-polarity form, southeast and down, demagnetized above 400 °C. (c) P in normal-polarity form, northwest and up. (d) AF demagnetization proved less effective at resolving P from the near-vertical remanence component. Temperatures are in °C, peak alternating fields are in mT.

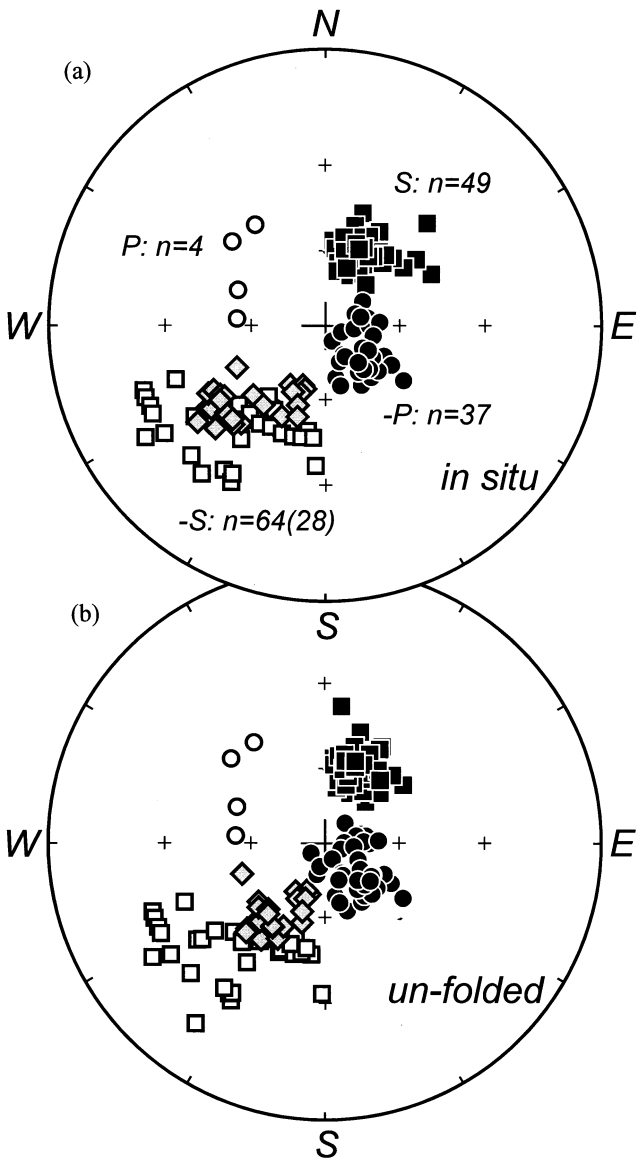


Figure 6. Equal-angle stereonet projections of (a) *in situ* and (b) unfolded remanence directions. Circles: remanence component P (P = normal polarity, −P = reverse polarity); squares and diamonds: remanence component S (S = normal, −S = reverse). The diamonds ($n = 28$) are a subset of the reverse-polarity S group ($n = 64$) for use in the fold test. Open (closed) symbols denote upward- (downward-) pointing remanences.

2.4 The magnetization mode of P

All P directions are given in Fig. 6(a) ($n = 41$). The reverse-polarity group ($n = 37$) is tightly clustered. The normal-polarity directions from the top of the Tosna section are few (4) and dispersed. Few of the P directions were from the neotectonic disturbance at Popovka and therefore unfolding (Fig. 6b) has virtually no effect on the groupings, and alters mean directions by only a few degrees (Table 1). The data are taken to have significance in unfolded form; mean directions are given in Fig. 7.

The normal- and reverse-polarity counterparts of P are in poor alignment and fail McFadden & McElhinny's (1990) reversal test ($P = 0.05$). We attribute this to the poor resolution of normal-polarity P directions during stepwise demagnetization. In Fig. 2 we have attempted to translate the stratigraphic positions of P remanence components in the three river sections onto a single magnetostratigraphic log (labelled Popovka–Tosna–Lava at the right-hand edge of the figure). In the combined log, horizons where P was observed are labelled with the name of the river section from which they were obtained. This compounded log provides a clearer indication of the stratigraphic ranges of P in reverse-polarity form (Tremadoc–Arenig–Llanvirn) and P in normal-polarity form (Llanvirn).

P always has the same polarity at the same stratigraphic horizon and is therefore presumed to be primary. This interpretation depends heavily on the observations of P at the top of the Tosna River section. We tested our interpretation against the Ordovician magnetostratigraphic log compiled by Trench, McKerrow & Torsvik (1991), which is based mainly on the work of Khramov, Rodionov & Komissarova (1965). The part of Trench *et al.*'s log covering the age range of the sampled stratigraphic section is given on the left-hand side of Fig. 2. The reference log has a wide reverse-polarity interval extending in time from the top of the Tremadoc to the middle part of the Llanvirn (*D. Murchisoni* zone). Supporting our

Table 1. Palaeomagnetic remanence directions and corresponding virtual geomagnetic pole positions. Arrows indicate geologically significant pole positions. ID = remanence component ID (P = primary, S = secondary; n, r and d refer to normal, reverse and dual polarity); Age = stratigraphic age (Arg = Arenig, Tr = Triassic); *D*, *I*, *n*, *R*, *k*, α_{95} = remanence declination, inclination, number of observations, resultant of *n* unit vectors, precision parameter, confidence limit; Lat., Long., *dp*, *dm* = pole latitude, longitude, confidence limits. All data are shown in both *in situ* and unfolded form. S_d is S_n combined with S_r.

ID	Polarity	Age	Tectonic	*D* (°)	*I* (°)	*n*	*R*	*k*	α_{95} (°)	Lat. (°N)	Long. (°E)	*dp*/*dn* (°/°)
P_r	R		*in situ*	124.5	71.3	37	36.4	55.9	3.2	33.8	64.1	4.9/5.6
	→	Arg	unfolded	130.4	73.1	37	36.2	44.9	3.6	34.7	59.1	5.7/6.4
P_n	N		*in situ*	303.1	−48.8	4	3.9	25.8	18.5	—	—	—
		Arg	unfolded				no change: as *in situ*					
S_n	N		*in situ*	25.8	56.4	49	48.2	62.2	2.6	61.8	162.9	2.7/3.8
		Tr	unfolded	23.9	56.8	49	48.4	78.8	2.3	62.9	165.3	2.4/3.3
S_r	R		*in situ*	223.7	−47.7	28	27.1	31.1	5.0	47.3	147.1	4.2/6.5
		Tr	unfolded	215.0	−56.1	28	27.6	66.3	3.4	57.8	150.5	3.5/4.9
S_d	N + R		*in situ*	33.0	53.6	77	74.8	35.2	2.8	56.5	155.5	2.7/3.9
	→	Tr	unfolded	28.0	56.7	77	75.9	67.9	2.0	61.3	159.3	2.1/2.9

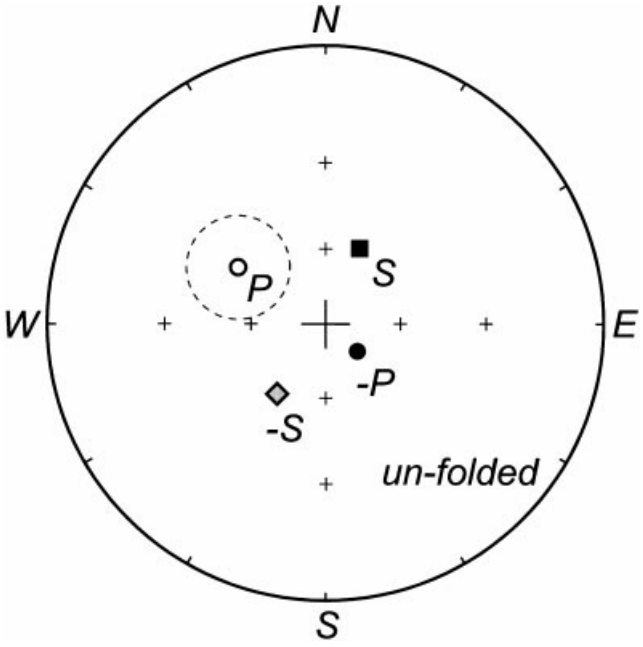


Figure 7. Mean directions for remanence components P (P = normal polarity, −P = reverse polarity) and S (S = normal, −S = reverse). The reverse-polarity direction for S (labelled −S) is the mean of the subset of directions used in the fold test ($n = 28$, Fig. 6). The cone of 95 per cent confidence on normal-polarity P is indicated. The cones of 95 per cent confidence in the other mean directions are of similar sizes to the symbols used. The mean directions are listed in Table 1.

argument for a primary mode for P, our magnetostratigraphy for P (on the right) mimics the 'global' magnetostratigraphy.

A primary mode for P is consistent with the genesis of its inferred host, magnetite. An early slightly reducing environment is indicated by the presence of glauconite. It is most likely that the magnetite was introduced at this early stage, either in biogenic or in diagenetic form. The later widespread chemical breakdown of glauconite to iron hydroxides in a near-surface position implies an oxidizing environment for much of the history of the limestone, making the late-stage genesis of magnetite unlikely. Strengthening the case for early magnetite, it is notable that P is often carried by the green limestone, the variety with the most primitive mineralogy.

3 COMPARISON WITH THE APW PATH FOR BALTICA

We have compared the apparent palaeomagnetic pole positions corresponding to S and P with the palaeomagnetic data compilation of Torsvik *et al.* (1996) for Vendian to Permian Baltica (Fig. 8). Torsvik *et al.* assigned numerical ages to stratigraphically dated rocks according to the timescale of Harland *et al.* (1990). We adjusted those numerical ages to conform with the new timescale for early and mid-Palaeozoic strata of Tucker & McKerrow (1995). The part of the palaeomagnetic data set relevant to the present study is the Vendian to Silurian segment. We therefore detail that subset in Table 2, together with the pole corresponding to P of this study. Those poles, labelled A to R, are indicated in Fig. 8.

We suspect a Mesozoic age for S and have therefore included Mesozoic data in our path. Rather than carry out a detailed review of available Mesozoic data we took as a starting point the selection of data in Van der Voo (1993) for stable Europe. In our analysis we retained only those data which had Van der Voo Q factors (Van der Voo 1988) of 4 or higher, leaving out data from poorly dated rocks and rocks with uncertain tectonic histories. These poles, numbered 1 to 23, are detailed in Table 3 and indicated in Fig. 8.

The pole corresponding to our presumed secondary remanence S ('S' in Fig. 8) falls on the southern edge of a group of Triassic poles, suggesting a Triassic age. The Mesozoic palaeomagnetic data for Europe describe a noisy arc and therefore only loosely constrain the age of S.

The pole corresponding to our primary remanence P (M in Table 2 and Fig. 8), from latest Tremadoc and Arenig rocks, lies to the northeast of a group of Ordovician results from southern Sweden (F to L). The linear age progression from the new result towards the southwest through the younger results ending at F confirms a primary mode for P.

4 A NEW APW PATH FOR BALTICA

The integrity of the tight swing in the Early Ordovician to Vendian part of the polar wander path, predicting high apparent rotation rates for Baltica, depends entirely on the palaeomagnetic results from the Fen Complex in southern

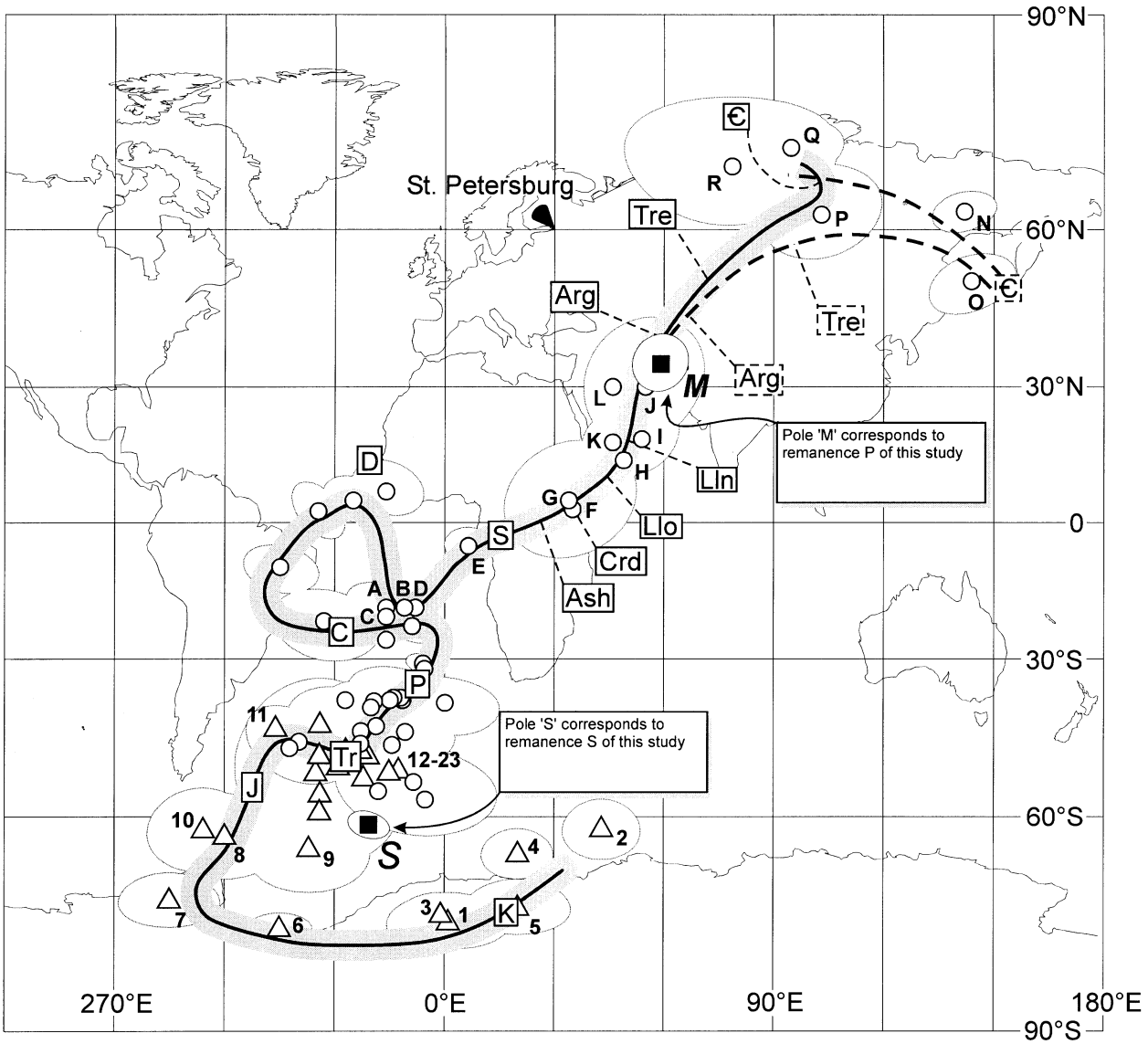


Figure 8. The new APW path for Baltica for 580–100 Ma (full line; Table 4). The path is based on the data selection of Torsvik *et al.* (1996) with the following changes: (1) addition of the new palaeomagnetic pole for remanence P of this study (P_r in Table 1); (2) removal of the Fen Complex poles which we suspect to be Permian/Triassic overprints (invert the Fen poles N and O); (3) adoption of the new timescale of Tucker & McKerrow (1995) for the early Palaeozoic (Table 2); and (4) addition of Mesozoic poles (triangles numbered 1–23; Table 3). Silurian to late Precambrian data (circles) are labelled A to R in accordance with Table 2 [M (square) is the pole corresponding to P of this study]. The pole marked S (square) corresponds to remanence S of this study (S_d in Table 1). Ages in the path are as follows: Є = dawn of the Cambrian (545 Ma); Tre = Tremadoc (495 Ma, Ordovician); Arg = Arenig (485 Ma); Lln = Llanvirn (470 Ma); Llo = Llandeilo (464 Ma); Crd = Caradoc (458 Ma); Ash = Ashgill (449 Ma); S = Silurian (443 Ma); D = Devonian (*c.* 415 Ma); C = carboniferous (363 Ma); P = Permian (290 Ma); Tr = Triassic (245 Ma); J = Jurassic (208 Ma); K = Cretaceous (146 Ma). The dashed line denotes the effect of the Fen Complex poles on the shape and time progression of the APW path. Smooth APW paths were generated digitally using the spherical spline-fitting routine of Jupp & Kent (1987) as implemented in the GMAP program of Torsvik & Smethurst (http:/www.geophysics.ngu.no/). 95 per cent error ovals on poles are indicated with dotted lines.

Norway (N and O, Table 2). A re-examination of the palaeomagnetic signature of the Fen Complex carbonatites by Meert & Torsvik (personal communication, 1997) leaves the question of remanence age open. Meert & Torsvik note the close correspondence between the Fen data (invert 'N' and 'O' in Table 2 and Fig. 8) and Permo-Triassic results (Fig. 8, Table 3). The presence of a major 50 km wide Permian igneous tract 10 km to the east (the Oslo Rift) adds to our concern regarding the Fen data. We have therefore generated a new APW path *excluding* the Fen Complex results and *including* the Early Ordovician P result (solid line in Fig. 8; Table 4). There is some uncertainty surrounding the radiometric age of Fen. New results suggest that Fen approaches the age of the Kola dykes in the palaeomagnetic data set (Table 2). Therefore, excluding the Fen Complex results from our data set may *not* result in a widening in the early Palaeozoic/latest Precambrian temporal gap in the APW record; we simply conclude that the gap is probably wider than we first thought (up to 100 Ma).

To highlight the geodynamic significance of removing the Fen Complex poles we compare our new APW path given by a solid line in Fig. 8 with the 'Fen' path given by a dashed line in Fig. 8 (see Table 4). The 'Fen' path is essentially a direct

Table 2. List of published palaeomagnetic data for Baltica for the period Vendian to Silurian. ID = pole letter in Fig. 8; Harland = ages according to Harland *et al.* (1990); Tucker = ages according to Tucker & McKerrow (1995); α_{95} = cone of 95 per cent confidence on the mean remanence direction; Q = sum of Van der Voo's (1988) reliability criteria satisfied (0 to 7); Pole Lat., Long. = virtual geomagnetic south pole latitude and longitude; Path Lat., Long. = position in the fitted APW path corresponding in time to the virtual geomagnetic south pole.

	Rock unit	Harland (Ma)	Tucker (Ma)	α_{95} (°)	Q	Pole Lat. (°N)	Pole Long. (°E)	Path Lat. (°N)	Path Long. (°E)	Reference
A	Ringerike Sandstone	420	←	9.1	7	−19	344	−17	345	Douglass (1988)
B	Gotland Dacker Limestone	425	←	2.0	4	−19	349	−20	349	Claesson (1979)
C	Gotland Follingbo Limestone	425	←	6.0	3	−21	344	−20	349	Claesson (1979)
D	Gotland Visby Limestone	428	←	5.1	5	−19	352	−19	351	Trench & Torsvik (1991)
E	Oslo Limestone	440	←	5.4	5	−5	7	−7	7	Bohm (1989)
F	Swedish Limestone I(N)	(459)→	456	13.4	5	3	35	3	32	Torsvik & Trench (1991a)
G	Vestergotland (N3)	(465)→	460	4.8	6	5	34	6	38	Torsvik & Trench (1991b)
H	Vestergotland (N1-N2 and R13)	(471)→	466	4.4	6	14	49	13	47	Torsvik & Trench (1991b)
I	Gullhøgen	(476)→	470	6.8	6	19	54	18	50	Tosvik *et al.* (1995c)
J	Swedish Limestones	(481)→	475	9.0	5	30	55	26	52	Perroud, Robardet & Bruton (1992)
K	Swedish Limestones I(R)	(481)→	475	5.1	6	18	46	26	52	Torsvik & Trench (1991a)
L	Swedish Limestones	(481)	475	2.2	5	30	46	26	52	Claesson (1978)
M	**St. Petersburg Limestones (remanence P)**	**(485)→**	**478**	**3.6**		**34.7**	**59.1**	**33**	**55**	**This study**
N	Fen Carbonate complex	(565)	←	3.0	3	63	142	56	143	Poorter (1972)
O	Fen Tinguates	(565)	←	6.4	4	51	144	56	143	Piper (1988)
P	Komagnes Dyke	(580)	←	4.5	3	63	103	69	94	Torsvik, Roberts & Siedlecka (1995a)
Q	Sredny Dyke	(580)	←	2.3	3	73	95	69	94	Torsvik *et al.* (1995a)
R	Sredny Peninsula Diabase dyke	(580)	←	5.0	3	70	79	69	94	Shipunov (1988)

Table 3. Mesozoic virtual geomagnetic pole positions for Baltica selected from the pole list of Van der Voo (1993). No. = pole number in Fig. 8. Reference for pole 1: Heller & Channell (1979); 2: Vincenz & Jalenska (1985); 3: Galbrun (1985); 4: Halvorsen (1989); 5: Ogg *et al.* (1991); 6: Hiijab & Tarling (1982); 7: Galbrun, Gabilly & Rasplus (1988); 8: Kadzialko-Hofmokl, Kruczyk & Westphal (1988); 9: Girdler (1968); 10: Girdler (1968); 11: Creer (1959); 12: Rother (1971); 13: Khramov (1975) (Pole 6.02), 14: Karmanova in Khramov (1971); 15: Khramov (1975) (Pole 6.10); 16: Khramov (1975) (Pole 6.16C); 17: Irving, Tanczyk & Hastie (1976) (Pole 8.189); 18: Khramov (1975) (Pole 6.09); 19: Biquand (1977); 20: Khramov (1975) (Pole 6.46); 21: Mulder (1972); 22: Khramov (1975) (Pole 6.08); 23: Khramov (1975) (Pole 6.43B).

No.	Rock unit	Q (1 to 7)	α_{95} (°)	Pole Lat. (°N)	Pole Long. (°E)	Age (Ma)
1	Münster Basin Ls., Ger.	5	4	−76	1	88–98
2	Dykes/Sills Spitzbergen	6	5	−62	43	100–115
3	Berriasian Stratotype, Fr.	6	3	−75	359	138–144
4	Hinlopenstr. Dol. Spitz.	6	4	−66	20	132–166
5	Krakow Upland Sed., Pol.	5	4	−74	20	160–169
6	Pliensbachian Sed., UK	4	2	−77	315	193–198
7	Toarcian Stratotype, Fr.	6	4	−73	285	187–193
8	Alsace Bajocian Sed., Fr.	7	7	−63	300	176–180
9	Liassic Volcanics, Fr.	4	7	−65	323	180–208
10	Keuper Volcanics, Fr	4	7	−62	294	208–230
11	Keuper Marls, UK	4	0.5	−44	314	208–230
12	Bunter & Musschelk, Ger.	5	15	−49	326	230–245
13	Yushatyr & Bukobai, Rus.	4	7	−48	337	208–230
14	Tuffs, Central Urals, Rus.	5	13	−48	333	230–245
15	Volga Clays, Rus.	4	10	−52	347	239–245
16	Blyumenthal Grp., Rus	4	11	−49	339	239–245
17	Varieg. Suite, Rus.	4	1	−51	331	239–245
18	Induan & Bashk, Stages Ukr.	4	3	−52	325	239–245
19	Buntsandstein, Fr.	6	5	−43	326	239–245
20	Romaskka & Redbeds, Rus.	5	4	−52	345	239–245
21	Kingscourt Redbeds, Irel.	4	7	−59	326	239–245
22	Serebryansk, Ukr.	5	12	−56	326	239–245
23	Vetluga Red Clays, Rus.	5	4	−53	338	239–245

Table 4. The statistically fitted APW path for Baltica for 580 to 100 Ma ('Without Fen'; solid line in Fig. 8). The early Palaeozoic/late Precambrian corner in the path controlled by the Fen Complex data (rejected in the present study) is given in the columns headed 'With Fen'. The parts of the path directly controlled by the Silurian to Late Precambrian data listed in Table 2 are underlined (see also right-hand column in this Table). The 478 Ma position in the path is controlled by the pole for remanence component *P* of this study (pole ID = 'M' in Table 2 and Fig. 8). There are two slightly different path positions for this time, the first (left) with the influence of Fen removed and the second (right) slightly influenced by the Fen Complex data.

Age (Ma)	Without Fen		With Fen		
	Lat. (°N)	Long. (°E)			
100	−69	32			
130	−71	12			
165	−74	20			
175	−79	355			
185	−78	304			
190	−73	292			
200	−63	302			
212	−50	310			
220	−46	317			
230	−48	329			
242	−51	333			
260	−48	337			
270	−45	340			
281	−40	346			
290	−36	353			
310	−24	358			
320	−22	356			
335	−23	348			
350	−24	335			
365	−23	318			
380	−15	310			
400	−2	322			
410	3	338			
415	−7	343			Input poles from Table 2
420	−17	345			A
425	−20	349			B, C
428	−19	351			D
440	−7	7			E
456	3	32			F
460	6	38			G
466	13	47			H
470	18	50			I
475	26	52	Lat. (°N)	Long. (°E)	J, K, L
478	32	53	33	55	M (remanence P, this study)
493	46	65	55	85	No poles
500	56	78	59	107	No poles
522	?	?	50	152	No poles
530	65	99	46	157	No poles
543	67	104	43	159	No poles
550	69	104	45	157	No poles
565	70	100	56	143	Fen Complex: N, O
580	71	95	69	94	Kola: P, Q, R

update of Torsvik *et al.*'s (1996) path, resembling it closely in shape but not in time progression. The updated time progression results from our adding the P pole and adopting the timescale of Tucker & McKerrow (1995). Compared with Torsvik *et al.*'s geodynamic interpretation, the revised time progression of the 'Fen' path has the effect of increasing rotation rates and shifting the peak rate forward in time into the Ordovician (dashed line, Fig. 9).

The new path is simpler than the 'Fen' path in shape, and therefore implies a simpler pattern of motions for Baltica (solid line, Fig. 9). First, high northerly and southerly latitudinal drift rates in the Vendian and Cambrian, respectively (20 cm yr^{-1} and 15 cm yr^{-1}), are reduced to virtually zero (solid line). Second, the aforementioned high apparent rotation rate for the Ordovician of 3.5° Ma^{-1} (dashed line) is significantly reduced to 2° Ma^{-1} (solid line).

Although removing data will almost always make the shape of polar wander paths simpler, we favour the more realistic geodynamic model for Baltica based on the new path. Although we have no other data to constrain APW in this time window, our understanding of early Palaeozoic palaeogeographies is not aided by the use of unreliable data. We reiterate that the use of the Fen data as primary poles becomes increasingly problematic geodynamically if Fen is found, as preliminary results suggest, to be of similar age to the Kola data. In that case Baltica would be constrained to move even more abruptly.

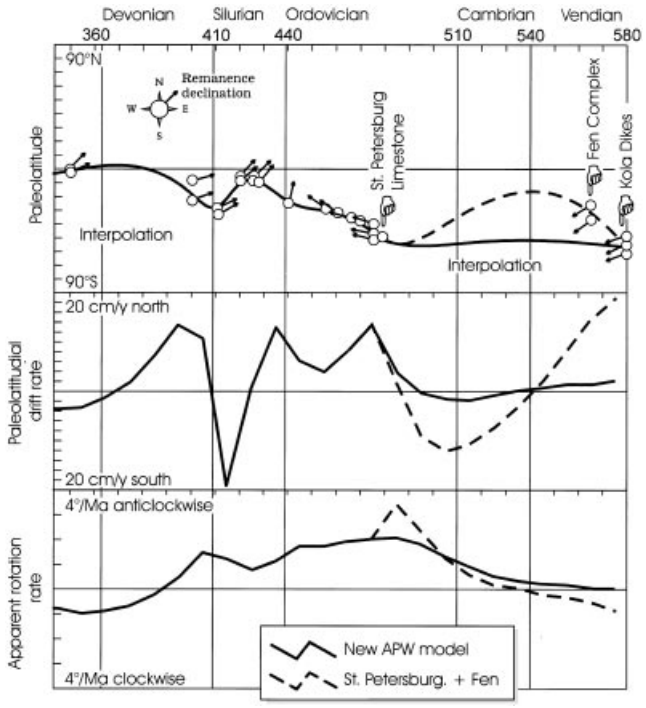


Figure 9. The pattern of motion of Baltica for Vendian to mid-Palaeozoic times. Top: change in palaeolatitude with time; middle: change in palaeolatitudinal drift rate; bottom: change in apparent rotation rate. Solid line: motion according to the new APW path of Fig. 8 (solid line). Dashed line: motion implied by the Fen Complex data (dashed line, Fig. 8). Arrows in the top display indicate the palaeonorth direction relative to present-day Baltica (remanence declination).

5 PALAEOGEOGRAPHY

Although the St. Petersburg result extends the APW record for Baltica backwards a few million years, the pole is similar in position to results from Swedish limestones (poles K and L in Fig. 8) and therefore does not alter the palaeomagnetic constraints on Early Ordovician palaeogeography. The reconstruction in Fig. 10, courtesy of Torsvik *et al.* (1996), holds for the new result. The figure, however, is altered to take account of the new data compilation for Siberia of Smethurst, Khramov & Torsvik (1998), which shifts Siberia 15° further north into an equatorial position. The new position increases the separation of Siberia and Baltica to *c.* 25° latitude, increasing the distance between the characteristic low-latitude trilobite forms of Siberia–Laurentia and the mid-latitude Baltica forms (Cocks & Fortey 1990). Geological information from Nikishin *et al.* (1996) and Pickering & Smith (1995) has been added. The longitudinal position of Laurentia is not constrained by the palaeomagnetic data, which leaves us free to slide Laurentia along lines of palaeolatitude. Laurentia may be moved in this way towards Gondwana (Fig. 10) to satisfy geological evidence as described by Dalziel (1997); however, we retain a traditional longitudinal position near to the present North Atlantic bordering continents to satisfy evidence for faunal exchange between them in the early Palaeozoic (McKerrow, Scotese & Brasier 1992).

Removing the late Vendian Fen Complex results from our data compilation makes a significant difference to the palaeoposition for Baltica at that time (Fig. 11). The 'with Fen' position straddles 30° south and the 'without Fen' position straddles 60° south (shaded). We have also changed the palaeolatitude of Siberia from the Torsvik *et al.* (1996) position to that of Smethurst *et al.* (1998). The change is a southerly shift in Siberia of 15°, bringing it closer to Gondwana. The position of Laurentia is taken from Torsvik *et al.* (1996).

Our adjustment in the position of Baltica takes it from low into intermediate to high southerly latitudes. This brings palaeomagnetic evidence, albeit interpolated between data, into alignment with the geological evidence offered by McKerrow *et al.* (1992). McKerrow *et al.* note a dominance of clastics in the Early Cambrian of Baltica, pointing to a temperate palaeolatitude. Low-latitude features such as archaeocyathan reefs, evaporites, dolomites and oolitic limestones are absent, in contrast to the then low-latitude continents of Laurentia and Siberia. To avoid overlap between continents at this time, Baltica is constrained to lie somewhere off the South American margin of Gondwana, if Gondwana is positioned according to Meert, Van der Voo & Ayub (1995). This position in relation to Gondwana, and the close proximity of Laurentia and Siberia (Fig. 11) are consistent with evidence for limited faunal exchange between Baltica, northwest Africa, Laurentia and Siberia in the early Palaeozoic (McKerrow *et al.* 1992).

6 CONCLUSIONS

Early Ordovician limestones south of St. Petersburg carry two dual-polarity remanence components, one secondary (S) and one primary (P). The secondary remanence corresponds to a palaeomagnetic North Pole at latitude 61.3°N, longitude 159.3°E (77 specimens, $dp/dm = 2.1°/2.9°$). Its age is taken to be Triassic from comparison with selected published poles. The polarity of S differs between parallel sampling traverses, indicating diachronous remagnetization of the limestones. We attribute S to chemical remanent magnetization produced by the breakdown of glauconite to goethite and haematite in the near-surface environment.

The primary remanence corresponds to a virtual geomagnetic south pole at latitude 34.7°N, longitude 59.1°E (37 specimens, $dp/dm = 5.7°/6.4°$). The remanence component is taken to be primary because it is resident in magnetite produced during early diagenesis and has a stratigraphically zoned reversal pattern. P has reversed polarity in latest Tremadoc and Arenig rocks and normal polarity in the uppermost Arenig/lowermost Llanvirn strata. This pattern matches other polarity records for the Ordovician period (Trench *et al.* 1991; Khramov *et al.* 1965; Gallet & Pavlov 1996; Torsvik *et al.* 1995b).

We construct an APW path for Vendian to Cretaceous time. This differs from earlier paths in (1) the inclusion of Mesozoic data, (2) the updating of early Palaeozoic pole ages according to the timescale of Tucker & McKerrow (1995), (3) the inclusion of the new Early Ordovician pole P, and (4) the exclusion of the Vendian Fen Complex results. The new pole pushes our knowledge of Phanerozoic APW back to the beginning of Ordovician times. Exclusion of the Fen Complex results removes the Vendian to earliest Ordovician corner in earlier paths. This simplifies the motion of Baltica, reducing latitudinal drift rates from 15/20 cm yr^{-1} to only a few centimetres per year and changing the clockwise followed by rapid anticlockwise rotation pattern into a single-sense gradual anticlockwise rotation. Exclusion of the Fen results leaves a

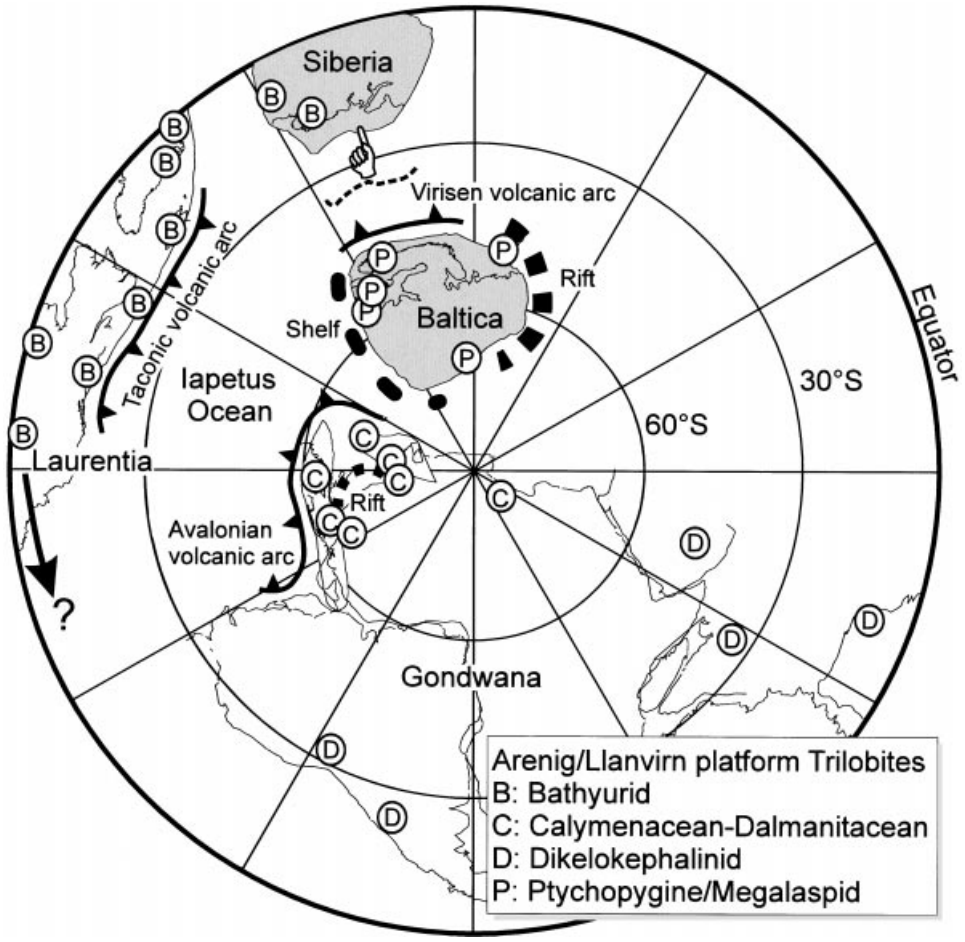


Figure 10. Plate reconstruction for 490–480 Ma. Baltica is positioned according to the St. Petersburg data (P). Siberia is positioned according to the APW path of Smethurst *et al.* (1998). Other continents are positioned after Torsvik *et al.* (1996) and references therein. Geological information is after Cocks & Fortey (1990), Nikishin *et al.* (1996) and Pickering & Smith (1995).



100 Ma gap in the APW record for Baltica, including all of the Cambrian period and a significant section of the Vendian. New Cambrian data are required to test our revised APW path and determine whether the Fen Complex results are to be exonerated as primary magnetizations.

We produce a revised plate configuration for Arenig time. The new result P confirms earlier models for the position and orientation of Baltica. Siberia is placed in low southerly latitudes according to the palaeomagnetic data compilation of Smethurst *et al.* (1998). We also generate a latest Vendian plate reconstruction. This map is significantly different from earlier palaeogeographies based on palaeomagnetic data (e.g. Torsvik *et al.* 1996). Exclusion of the Fen Complex data takes Baltica out of a low-latitude position and into a mid- to high-latitude position. This is consistent with sedimentological and biogeographical data

Figure 11. Plate reconstruction for Vendian/Cambrian time (*c.* 550 Ma). Baltica is positioned according to the new APW path of this study (excluding the Fen Complex data, Table 4). Siberia is positioned according to the APW path of Smethurst *et al.* (1998). Gondwana is reconstructed and positioned according to Meert *et al.* (1995). Laurentia is positioned according to Torsvik *et al.* (1996). T = tilloid; E = evaporite. Geological details are after McKerrow *et al.* (1992), Nikishin *et al.* (1996) and Pickering & Smith (1995).

(McKerrow *et al*. 1992). The position of Siberia is relocated 15° towards the south to a position nearer to Gondwana.

ACKNOWLEDGMENTS

We thank an anonymous reviewer for helpful comments and suggestions on an earlier version of the paper. This work was funded by the Norwegian Research Council and the Geological Survey of Norway.

REFERENCES

Biquand, D., 1977. Paléomagnétisme de la formation des 'grès-à-Voltzia' (Buntsandstein supérieur) du Massif des Vosges (France), *Can. J. Earth Sci.,* **14,** 1490–1514.

Bøhm, V., 1989. Vergleichende palaeomagnetische untersuchungen an verscheidenen Gesteinen de mitteleuropeischen Variszikums, *PhD thesis,* Ludwig-Maximilians University, Munich.

Claesson, K.C., 1978. Swedish Ordovician limestones: problems in clarifying their directions of magnetisation, *Phys. Earth planet. Inter.,* **16,** 65–72.

Claesson, K.C., 1979. Early Palaeozoic geomagnetism of Gotland, *Geol. Foren. Stockholm Forh.,* **101,** 149–155.

Cocks, L.R.N. & Fortey, R.A., 1990. Biogeography of Ordovician and Silurian faunas, in *Palaeozoic Palaeogeography and Biogeography,* eds McKerrow, W.S. & Scotese, C.R, *Geol. Soc. Mem.,* **12,** 97–104.

Creer, K.M., 1959. AC demagnetisation of unstable Triassic Keuper marls from SW England, *Geophys. J. R. astr. Soc.,* **2,** 261–275.

Dalziel, I.W.D., 1997. Neoproterozoic-Palaeozoic geography and tectonics: Review, hypothesis, environmental speculation, *Geol. Soc. Am. Bull.,* **109,** 16–42.

Douglass, D.N., 1988. Palaeomagnetics of Ringerike Old Red Sandstone and related rocks, southern Norway: implications for pre-Carboniferous separation of Baltica and British terranes, *Tectonophysics,* **148,** 11–27.

Galbrun, B., 1985. Magnetostratigraphy of the Berriasian stratotype section (Berrias, France), *Earth planet. Sci. Lett.,* **74,** 130–136.

Galbrun, B., Gabilly, J. & Rasplus, L., 1988. Magnetostratigraphy of the Toarcian stratotype section at Thouars and Airvault (Deux-Sèvres, France), *Earth planet. Sci. Lett.,* **87,** 453–462.

Gallet, Y. & Pavlov, V., 1996. Magnetostratigraphy of the Moyero river section (north-western Siberia): constraints on geomagnetic reversal frequency during the early Palaeozoic, *Geophys. J. Int.,* **125,** 95–105.

Girdler, R.W., 1968. A palaeomagnetic investigation of some Late Triassic and Early Jurassic volcanic rocks from the northern Pyrenees, *Ann. Geophys.,* **24,** 1–14.

Halvorsen, E., 1989. A palaeomagnetic pole position of Late Jurassic/Early Cretaceous dolerites from Hinlopenstretet, Svalbard, and its tectonic implications, *Earth planet. Sci. Lett.,* **94,** 398–408.

Harland, W.B., Armstrong, R.L., Cox, A.V., Craig, L.E., Smith, A.G. & Smith, D.G., 1990. *A Geological Time Scale,* Cambridge University Press, Cambridge.

Heller, F. & Channell, J.E.T., 1979. Palaeomagnetism of Upper Cretaceous limestones from the Münster Basin, Germany, *J. Geophys.,* **46,** 413–427.

Hijab, B.R. & Tarling, D.H., 1982. Lower Jurassic palaeomagnetic results from Yorkshire, England, and their implications, *Earth planet. Sci. Lett.,* **60,** 147–154.

Irving, E., Tanczyk, J. & Hastie, J., 1976. *Catalogue of Paleomagnetic Directions and Poles,* 4th issue, *Mesozoic results 1954–1975,* Geomagn. Ser., Vol. 6, Geomagn. Serv. Canada, Ottawa.

Jupp, P.E. & Kent, J.T., 1987. Fitting smooth paths to spherical data, *Appl. Stat.,* **36,** 34–36.

Kadzialko-Hofmokl, M. Kruczyk, J. & Westphal, M., 1988. Palaeomagnetism of Jurassic sediments from the western border of the Rheingraben, Alsace (France), *J. Geophys.,* **62,** 102–108.

Khramov, A.N., 1971. *Paleomagnetic Directions and Poles,* Issue no. 1, Moscow: Soviet Geophysical Committee of the Acad. Sci. USSR, World Data Center B.

Khramov, A.N., 1975. *Paleomagnetic Directions and Poles,* Issue no. 3, Moscow: Soviet Geophysical Committee of the Acad. Sci. USSR, World Data Center B.

Khramov, A.N., 1984. *Paleomagnetic Directions and Poles,* Summary catalogue no. 1, Moscow: Soviet Geophysical Committee of the Acad. Sci. USSR, World Data Center B.

Khramov, A.N. & Rodionov, V.P., 1977. Problems of the Early and Middle Palaeozoic Laurasia in the light of palaeomagnetic data, in *Paleomagnetizm i voprosy tektoniki plit.,* pp. 108–140, VNIGRI, Leningrad.

Khramov, A.N., Rodionov, V.P. & Komissarova, R.A., 1965. *New data on the Palaeozoic History of the Geomagnetic Field in the USSR,* Trans. Hope, E.R. (Defence Research Board, Canada, T 460 R) from Nastoyashcheye i Prochloye Magnitnogo Polia Zemli, pp. 206–213, Nauka, Moscow.

McFadden, P.L. & Jones, D.L., 1981. The fold test in palaeomagnetism, *Geophys. J. R. astr. Soc.,* **67,** 53–58.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.,* **103,** 725–729.

McKerrow, W.S., Scotese, C.R. & Brasier, M.D., 1992. Early Cambrian reconstructions, *J. geol. Soc. Lond.,* **149,** 599–606.

Meert, J.G., Van der Voo, R. & Ayub, S., 1995. Palaeomagnetic investigation of the Neoproterozoic Gagwe lavas and Mbozi complex, Tanzania and the assembly of Gondwana, *Precambrian Res.,* **74,** 225–244.

Mulder, F.G., 1972. Paleomagnetic results from Irish Carboniferous and Triassic rocks, *Sci. R. Dublin Soc.,* **A4,** 343–349.

Nikishin, A.M. *et al.,* 1996. Late Precambrian to Triassic history of the East European Craton: dynamics of sedimentary basin evolution, *Tectonophysics,* **268,** 23–63.

Obut, A.M., Sennikov, N.V. & Dmitrovskaya Ju.E., 1991. Graptolite complexes in the Ordovician of the Leningrad region, *Izv. Acad. Sci. USSR, ser. geol.,* **N4,** 14–20 (in Russian).

Ogg, J.G., Steiner, M.B., Wieczorek, J. & Hoffmann, M., 1991. Jurassic magnetostratigraphy: 4. Early Callovian through Middle Oxfordian of the Krakow Uplands (Poland), *Earth planet. Sci. Lett.,* **104,** 488–504.

Perroud, H., Robardet, M. & Bruton, D.L., 1992. Palaeomagnetic constraints upon the palaeogeographic position of the Baltic Shield in the Ordovician, *Tectonophysics,* **201,** 97–120.

Pickering, K.T. & Smith, A.G., 1995. Arcs and backarc basins in the early Palaeozoic Iapetus Ocean, *The Island Arc,* **4,** 1–67.

Piper, J.D.A., 1988. Palaeomagnetism of (late Vendian-earliest Cambrian) minor alkaline intrusions, Fen Complex, south-east Norway, *Earth planet. Sci. Lett.,* **90,** 422–430.

Poorter, R.P.E., 1972. Preliminary palaeomagnetic results from the Fen Carbonatite complex, S. Norway, *Earth planet. Sci. Lett.,* **17,** 194–198.

Rother, K., 1971. Gesteins- und paläomagnetische Untersuchungen an Gesteinsproben vom Territorium der DDR aus dem Präkambrium bis zum Teriär und Fölgerungen für die Veränderungen des geomagnetischen Hauptfeldes sowie für geologisch-geotektonische Iinerpretationsmöglichkeiten, *Veröffentl,* **5,** Zentralinstitut Physik der Erde, Potsdam.

Selivanova, V.A. & Kofman, V.C., 1971. *Geology of the USSR,* Vol. 1, Nedra, Moscow (in Russian).

Shikolukov Ju.A. & Bibikova, E.V., 1986. *Methods of Isotopic Geology and the Geochronological Scale,* Nauka, Moscow (in Russian).

Shipunov, S.V., 1988. Stratigraphic consistency criterion on Palaeomagnetic studies. Izvestiya, Academy of Sciences, USSR. *Phys. Solid. Earth,* **24** (5), 421–425.

Smethurst, M.A., 1991. A practical suggestion regarding the use of Scandinavian and Russian palaeomagnetic data to determine the palaeo-position of Baltica in Ordovician time, *Tectonophysics,* **201,** 65–73.

Smethurst, M.A., Khramov, A.N. & Torsvik, T.H., 1998. The Neoproterozoic and Palaeozoic palaeomagnetic data for the Siberian platform: from Rodinia to Pangea, *Earth Sci. Rev.,* **43,** 1–25.

Torsvik, T.H. & Trench, A., 1991a. The Lower-Middle Ordovician of Scandinavia: southern Sweden 'revisited', *Phys. Earth planet. Inter.,* **65,** 283–291.

Torsvik, T.H. & Trench, A., 1991b. Ordovician magnetostratigraphy: Llanvirn–Caradoc limestones of the Baltic Platform, *Geophys. J. Int.,* **107,** 171–184.

Torsvik, T.H., Smethurst, M.A., Briden, J.C. & Sturt, B.A., 1990. A review of Palaeomagnetic data from Europe and their palaeogeographical implications, in *Palaeozoic Palaeogeography and Biogeography,* eds McKerrow, W.S. & Scotese, C.R., *Geol. Soc. Mem.,* **12,** 25–41.

Torsvik, T.H. Smethurst, M.A., Van der Voo, R., Trench, A., Abrahamsen, N. & Halvorsen, E., 1992. Baltica. A synopsis of Vendian-Permian palaeomagnetic data and their paleotectonic implications, *Earth Sci. Rev.,* **33,** 133–152.

Torsvik, T.H., Roberts, D. & Siedlecka, A., 1995a. Palaeomagnetic data from sedimentary rocks and dolerite dykes, Kildin Island, Rybachi, Sredni and Varanger Peninsulas, NW Russia and NE Norway: A review, *Nor. geol. unders. Spec. Publ.,* **7,** 315–326.

Torsvik, T.H., Tait, J., Moralev, V.M., McKerrow, W.S., Sturt, B.A. & Roberts, D., 1995b. Ordovician palaeogeography of Siberia and adjacent continents, *J. Geol. Soc. Lond.,* **152,** 279–287.

Torsvik, T.H., Trench, A., Lohmann, K.C. & Dunn, S., 1995c. Lower Ordovician reversal asymmetry: An artefact of remagnetisation or non-dipole field disturbance?, *J. geophys. Res.,* **100,** 17 885–17 898.

Torsvik, T.H., Smethurst, M.A., Meert, J.G., Van der Voo, R., McKerrow, W.S., Brasier, M.D., Sturt, B.A. & Walderhaug, H.J., 1996. Continental break-up and collision in the Neoproterozoic and Palaeozoic—A tale of Baltica and Laurentia, *Earth Sci. Rev.,* **40,** 229–258.

Trench, A. & Torsvik, T.H., 1991. The Lower Palaeozoic apparent polar wander path for Baltica: palaeomagnetic data from Silurian limestones of Gotland, Sweden, *Geophys. J. Int.,* **107,** 373–379.

Trench, A., McKerrow, W.S. & Torsvik, T.H., 1991. Ordovician magnetostratigraphy: a correlation of global data, *J. Geol. Soc. Lond.,* **148,** 949–957.

Tucker, R.D. & McKerrow, W.S., 1995. Early Palaeozoic chronology: a review in light of new zircon ages from Newfoundland and Britain, *Can. J. Earth Sci.,* **32,** 368–379.

Van der Voo, R., 1988. Palaeozoic palaeogeography of North America, Gondwana, and intervening displaced terranes: comparisons of palaeomagnetism with palaeoclimatology and biogeographical patterns, *Geol. Soc. Am. Bull.,* **100,** 311–324.

Van der Voo, R., 1993. *Palaeomagnetism of the Atlantic, Tethys and Iapetus Oceans,* Cambridge University Press, New York, NY.

Vincenz, S.A. & Jalenska, M., 1985. Paleomagnetic investigations of Mesozoic and Palaeozoic rocks from Svalbard, *Tectonophysics,* **114,** 163–180.